Jonas Mehtali

PhD Student in Computer-Assisted Interventions

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Languages: French (native), German (native), English (C2, LanguageCert Level 3, 2022)

SUMMARY

PhD student at ICube (UMR 7357), University of Strasbourg, working with Caroline Essert and Juan Verde on automatic multi-needle planning for percutaneous thermal ablation of liver tumors: multi-needle trajectory optimization, GPU-accelerated ablation simulation, and an ergonomic planning interface. Funded through the ITI HealthTech doctoral program, ED269 MSII.

EDUCATION

PhD in Computer-Assisted Interventions *2024 - present*
ICube, University of Strasbourg / IHU Strasbourg. Supervisors: Pr. Caroline Essert, Dr. Juan Verde.

MSc in Computer Science, Image and 3D *2022 - 2024*
University of Strasbourg (M1 12.9/20, M2S1 13.7/20)

BSc in Computer Science (CMI engineering track) *2019 - 2022*
University of Strasbourg (14.3/20)

Institut Sainte-Jeanne d'Arc, Mulhouse *2016 - 2019*
Baccalaureat General (15.4/20), Allgemeine Hochschulreife (German Abitur, 1.2)

RESEARCH EXPERIENCE

PhD Researcher, ICube (IMAGeS) and IHU Strasbourg *Sep 2024 - present*

- Multi-needle trajectory optimization for thermal ablation, comparing exhaustive and learning-based methods with respect to vital anatomical structures
- GPU multi-resolution ablation simulation (HEAT, results under 1 s) and chained Neural Cellular Automata estimators (C-NCA, 476 fps)
- Intraoperative replanning under breathing motion; usability studies with clinical experts

End-of-Studies Internship, ICube (IMAGeS) and IHU Strasbourg *Jan - Aug 2024*

- Real-time thermal ablation simulation with networked GPU compute; mixed-reality co-planning across devices; custom networking protocol with modified Huffman compression

Research Internship, TU Darmstadt (MEC-Lab) and ICube (IMAGeS) *Jun - Sep 2023*

- CryoTrack cryoablation planning and navigation: breathing-cycle synchronization, EM tracking, real-time needle trajectory correction; phantom studies at CHU Strasbourg; led to a MICCAI 2024 publication

Research Camp, University of Oviedo, Spain *May - Jul 2022*

- Full research project from hypothesis to publication; co-authored the resulting PeerJ paper on typeface usability

Research Internship, EOST, University of Strasbourg *Jun - Jul 2020*

- Landslide monitoring software (Python/Matplotlib) for the La Valette site, later taken over by an EOST student

PUBLICATIONS AND PATENT

- **J. Mehtali**, J. Verde, C. Essert. C-NCA: Chained Neural Cellular Automata for Fast and Accurate Thermal Ablation Estimation. MICCAI 2025, Seoul. DOI 10.1007/978-3-032-04965-0_7
- **J. Mehtali**, J. Verde, C. Essert. HEAT: High-Efficiency Simulation for Thermal Ablation Therapy. IJCARS 2025, 20(6):1135-1143. DOI 10.1007/s11548-025-03350-z
- H.J. Krumb, **J. Mehtali**, J. Verde, A. Mukhopadhyay, C. Essert. Cryotrack: Planning and Navigation for Computer Assisted Cryoablation. MICCAI 2024, Marrakesh. DOI 10.1007/978-3-031-72089-5_10
- S. Vecino, **J. Mehtali**, J. de Andres, et al. How does serif vs sans serif typeface impact the usability of e-commerce websites? PeerJ Computer Science, 2022. DOI 10.7717/peerj-cs.1139
- Patent: Method and apparatus for determining a thermal ablation volume (J. Mehtali, C. Essert, J. Verde)

SKILLS

Languages	Python, C/C++, CUDA
Medical imaging	3D Slicer, PyTorch, VTK, OpenIGTLink, medical image processing and segmentation
Graphics and XR	OpenGL, Unity, Godot, Blender, Maya, mixed reality (Meta Quest)
Other	GPU computing, LLM tooling and automation, project management, scientific writing

SHORT POSITIONS

- Caterer and event staff, Service de la Vie Universitaire, University of Strasbourg (2022 - 2024)
- General agent and village shop sales and accounting, Town of Niffer, Haut-Rhin (2020 - 2022)
- IT maintenance intern, Concept Office, Mulhouse (2018)